

Chapter 2: Methodology

2.1 Introduction

This chapter presents the comprehensive methodology employed in developing an Explainable AI (XAI) framework for brain tumor segmentation using 3D Convolutional Neural Networks (CNNs). The primary objective is to transform an opaque deep learning model into a transparent system that not only achieves high segmentation accuracy but also provides clinically meaningful insights into the decision-making process. This work addresses the critical challenge of removing the "black box" problem in medical imaging AI by implementing six complementary XAI techniques and evaluating them against ground truth annotations[1].

The chapter is organized as follows: Section 2.2 provides a brief overview of the end-to-end framework architecture, Section 2.3 introduces the XAI methodology ecosystem, Section 2.4 details the SegResNet model implementation, Section 2.5 describes the complete pipeline workflow, and Section 2.6 summarizes the quantitative evaluation strategy encompassing both traditional segmentation metrics and novel XAI-specific evaluation metrics.

Key Contributions of This Chapter:

- Implementation of six XAI methods adapted for 3D volumetric medical imaging
 - Development of four quantitative XAI evaluation metrics (Pointing Game, Saliency Coverage, Saliency IoU, Weighted Dice)
 - Introduction of a novel Weighted Dice metric for continuous saliency evaluation
 - Integration of MC Dropout for uncertainty quantification
 - Complete evaluation pipeline bridging model prediction accuracy with explainability validity
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2.2 End-to-End Framework Overview

2.2.1 High-Level Architecture

The framework follows a modular, sequential pipeline designed to separate concerns and enable independent validation of each component:

Figure 1: End-to-end XAI framework for brain tumor segmentation. The pipeline progresses from data preprocessing through 3D segmentation to six parallel XAI interpretation methods, culminating in VR visualization.

Core Pipeline Stages:

1. **Data Acquisition & Preprocessing** — BraTS 2023 dataset (~1,250 patients) with 4 MRI modalities (T1, T1-contrast, T2, T2-FLAIR) normalized and cropped to $160 \times 160 \times 160$ voxels
2. **3D Segmentation Model** — SegResNet architecture trained on 1,000 patients to produce 3-channel tumor predictions (Whole Tumor, Tumor Core, Enhancing Tumor)
3. **XAI Method Suite** — Six parallel techniques generating spatial saliency maps explaining model decisions
4. **Metric Evaluation** — Quantitative assessment against ground truth (test set: ~125 patients)
5. **VR Visualization** — Immersive 3D representation in SlicerVR for clinical decision support

2.2.2 Dataset Overview

BraTS 2023 Dataset Composition:

Component	Details
Total Patients	~1,250
Input Modalities	T1-native, T1-Gadolinium (contrast), T2-weighted, T2-FLAIR
Spatial Resolution	1 mm × 1 mm × 1 mm (interpolated to 160 × 160 × 160 voxels)
Tumor Classes	3: Whole Tumor (WT), Tumor Core (TC), Enhancing Tumor (ET)
Data Split	1,000 training, 125 validation, 125 test
Preprocessing	Z-score normalization (per-channel, non-zero voxels only), foreground cropping

Label Definition (BraTS 2023 conventions):

Whole Tumor (WT) = {1, 2, 3} (all non-background)
 Tumor Core (TC) = {1, 3} (necrosis + enhancing)
 Enhancing Tumor (ET) = {3} (actively enhancing)

2.2.3 Implementation Technologies

Component	Technology	Purpose
Deep Learning Framework	PyTorch 2.0+	GPU-accelerated neural network training and inference
Medical Imaging Library	MONAI	Pre-built medical imaging transforms and architectures
Model Architecture	SegResNet (MONAI)	3D residual encoder-decoder for volumetric segmentation
XAI Implementation	Custom PyTorch hooks	Gradient-based and perturbation-based explanations
Visualization	3D Slicer / SlicerVR	Clinical-grade volumetric rendering and VR display
Experiment Tracking	Weights & Biases	Reproducible model training logs and metrics

2.3 Explainable AI (XAI) Framework

2.3.1 Motivation for XAI

Medical AI systems must be interpretable to gain clinical acceptance. A model with 90% accuracy is only useful if clinicians can understand *where* and *why* the model is making decisions[2]. This project implements six complementary XAI techniques, each answering different aspects of the question: **"Where is the model looking when it makes a segmentation decision, and does that match the actual tumor?"**

2.3.2 Six XAI Methods

The framework implements six explainability techniques spanning gradient-based, perturbation-based, and uncertainty-based approaches:

2.3.2.1 Grad-CAM (Gradient-weighted Class Activation Mapping)

Concept: Computes per-feature-map importance weights using gradients flowing into the bottleneck layer.

Mathematical Formulation:

For target class c and feature map k :

$$\alpha_k^c = \frac{1}{Z} \sum_{d,h,w} \frac{\partial y^c}{\partial A_{d,h,w}^k}$$

$$L_{\text{GradCAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c \cdot A^k \right)$$

Where:

- y^c = spatially-averaged output logit for class c
- A^k = activation of feature map k at the bottleneck ($20 \times 20 \times 20$ resolution)
- Z = total spatial elements
- ReLU retains only positive contributions

Properties:

- **Resolution:** Coarse ($20^3 \rightarrow$ upsampled to 160^3)
- **Class-Specific:** Yes (separate heatmap per class)
- **Sharpness:** Low (blurry blob due to upsampling)
- **Computational Cost:** Single forward+backward pass
- **Target Layer:** Bottleneck (Level 4, 256 channels)

Output: Three Nifti files per patient (WT, TC, ET heatmaps)

Reference: Selvaraju et al., "Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization", IJCV 2020[3]

2.3.2.2 Guided Backpropagation (GBP)

Concept: Modifies the standard backpropagation algorithm by adding an additional gating mechanism at every ReLU layer during the backward pass.

Mathematical Formulation:

Standard ReLU backprop: $\frac{\partial R}{\partial x} = \frac{\partial R}{\partial f} \cdot \mathbb{1}(f > 0)$

Guided Backprop: $\frac{\partial R}{\partial x} = \frac{\partial R}{\partial f} \cdot \mathbb{1}(f > 0) \cdot \mathbb{1}\left(\frac{\partial R}{\partial f} > 0\right)$

Input gradient (saliency):

$$S(d, h, w) = \max_c \left| \frac{\partial \text{Score}}{\partial x_{c,d,h,w}} \right|$$

Properties:

- **Resolution:** Full (160^3)
- **Class-Specific:** No
- **Sharpness:** High (sharp pixel-level detail)
- **Computational Cost:** Single backward pass with modified hooks
- **Interpretation:** "Which individual voxels drive the score?"

Output: Three NIfTI files per patient (WT, TC, ET saliency maps)

Reference: Springenberg et al., "Striving for Simplicity: The All Convolutional Net", ICLR 2015

2.3.2.3 Guided Grad-CAM

Concept: Combines the class-discriminative power of Grad-CAM with the pixel-level detail of Guided Backpropagation through element-wise multiplication.

Mathematical Formulation:

$$\text{Guided Grad-CAM} = \text{GBP} \odot \text{upsample}(\text{Grad-CAM})$$

Where $\odot$ denotes element-wise multiplication.

Properties:

- **Resolution:** Full (160^3)
- **Class-Specific:** Yes
- **Sharpness:** High (best of both methods)

- **Theoretical Basis:** Combines spatial localization (Grad-CAM) with feature attribution (GBP)

Output: Three NIfTI files per patient (WT, TC, ET maps)

2.3.2.4 Layer-wise Relevance Propagation (LRP)

Concept: Propagates the output prediction score backward through the network, distributing relevance based on the magnitude of activations and weights at each layer.

Mathematical Formulation (Input $\times$ Gradient Proxy):

The exact ϵ -LRP rule is computationally expensive for large networks. We use the mathematically equivalent Input $\times$ Gradient approximation:

$$R_i \approx x_i \cdot \frac{\partial f(x)}{\partial x_i}$$

Where:

- x_i = input voxel intensity
- $f(x)$ = model output score for target class
- R_i = relevance score for voxel i

Properties:

- **Resolution:** Full (160^3)
- **Class-Specific:** Yes
- **Interpretation:** "Which inputs contributed most to the final prediction?"
- **Computational Basis:** Deep Taylor Decomposition / Conservation Principle

Output: Three NIfTI files per patient (WT, TC, ET relevance maps)

Reference: Bach et al., "Layer-wise Relevance Propagation: An Overview", LNCS 2015

2.3.2.5 Occlusion Sensitivity

Concept: Tests model reliance on specific regions by systematically occluding (masking) spatial patches and recording the resulting drop in prediction confidence.

Mathematical Formulation:

$$S_c(i, j, k) = f_c(x) - f_c(x_{\text{occluded}})$$

Where:

- $f_c(x)$ = original prediction score for class c
- $f_c(x_{\text{occluded}})$ = score after masking a 16^3 region centered at (i, j, k)
- $S_c > 0$ indicates the region was important

Properties:

- **Resolution:** Coarse (20^3 sliding window $\rightarrow$ upsampled to 160^3)
- **Class-Specific:** Yes
- **Model-Agnostic:** Yes (requires only forward passes)
- **Computational Cost:** High (~ 1000 forward passes per patient)
- **Window Size:** $16 \times 16 \times 16$ voxels
- **Stride:** 8 voxels

Output: Three NIfTI files per patient (WT, TC, ET sensitivity maps)

Reference: Zeiler & Fergus, "Visualizing and Understanding Convolutional Networks", ECCV 2014

2.3.2.6 Monte Carlo Dropout (MC Dropout)

Concept: Quantifies model uncertainty by performing multiple stochastic forward passes with dropout layers active at inference time, treating the ensemble variance as uncertainty.

Mathematical Formulation:

Perform $N = 20$ forward passes with dropout enabled:

$$\bar{p}(x) = \frac{1}{N} \sum_{n=1}^N p_n(x) \quad (\text{ensemble mean})$$

$$\sigma^2(x) = \frac{1}{N} \sum_{n=1}^N (p_n(x) - \bar{p}(x))^2 \quad (\text{per-voxel variance/uncertain})$$

Properties:

- **Output:** Two maps per class (mean prediction + uncertainty)
- **Interpretation:** Low variance = confident, high variance = uncertain
- **No Retraining:** Uses existing model with dropout enabled
- **Boundary Behavior:** "Boundary glow" effect (uncertainty concentrates at tumor edges)

Output: Six NIFTI files per patient (mean predictions + uncertainty maps for WT, TC, ET)

Reference: Gal & Ghahramani, "Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning", ICML 2016[4]

2.3.3 Comparison of XAI Methods

Method	Resolution	Class-Specific?	Sharpness	Cost
Grad-CAM	Coarse (20 ³)	Yes	Low	Low
Guided Backprop	Full (160 ³)	No	High	Low
Guided Grad-CAM	Full (160 ³)	Yes	High	Low
LRP (Input × Grad)	Full (160 ³)	Yes	High	Low
Occlusion	Coarse (20 ³)	Yes	Low	High
MC Dropout	Full (160 ³)	Yes	N/A	Medium

Table 1: Comparison of XAI methods across key dimensions.

2.4 SegResNet Model Implementation

2.4.1 Architecture Overview

SegResNet is a 3D encoder-decoder architecture with residual blocks, specifically designed for volumetric medical image segmentation. The model processes entire brain MRI volumes without sliding windows during training, enabling global context modeling.

2.4.2 Network Architecture

Encoder (Compression Path):

The encoder progressively downsamples the input volume across 4 levels using strided convolutions (stride = 2), extracting increasingly abstract features:

Level	Filters	Spatial Res.	Description
Input	4	$160 \times 160 \times 160$	4 MRI modalities
Level 1	32	$160 \times 160 \times 160$	Low-level edge/texture features
Level 2	64	$80 \times 80 \times 80$	Mid-level structural patterns
Level 3	128	$40 \times 40 \times 40$	High-level semantic features
Level 4 (Bottleneck)	256	$20 \times 20 \times 20$	Deepest abstraction

Table 2: SegResNet encoder architecture (init_filters = 32).

Filter Progression: $\text{init_filters} \times 2^{(\text{level}-1)}$

Bottleneck Features: At Level 4, the 256-channel feature maps (20^3 resolution) capture the entire brain tumor context in a highly compressed form. This is the target layer for Grad-CAM analysis.

Decoder (Expansion Path):

The decoder mirrors the encoder but uses upsampling to restore spatial resolution:

- Level 4 → Level 3: Upsample $20^3 \rightarrow 40^3$, concatenate Level 3 encoder features (skip connection)
- Level 3 → Level 2: Upsample $40^3 \rightarrow 80^3$, concatenate Level 2 encoder features
- Level 2 → Level 1: Upsample $80^3 \rightarrow 160^3$, concatenate Level 1 encoder features
- Level 1 → Output: 160^3 with 3 channels (WT, TC, ET)

Skip Connections: Direct feature concatenation between matching encoder-decoder levels preserves fine-grained spatial information and gradients during training.

2.4.3 Residual Blocks

Every layer in SegResNet is built from residual blocks (ResBlocks). Instead of learning a direct mapping $f(x) \approx y$, ResBlocks learn the *difference*:

$$\text{ResBlock}(x) = x + f(x)$$

Where $f(x)$ is the output of: GroupNorm → ReLU → Conv3D(3^3) → GroupNorm → ReLU → Conv3D(3^3)

Benefits of ResBlocks:

1. **Gradient Flow:** Residual connections provide direct gradient paths, solving the vanishing gradient problem
2. **Interpretability:** Each block learns an interpretable "correction" rather than an opaque transformation
3. **Training Stability:** Enables training of very deep networks without degradation

2.4.4 Model Configuration

Hyperparameter	Value
Architecture	SegResNet (MONAI)
Spatial Dimensions	3D
Input Channels	4 (T1, T1-contrast, T2, T2-FLAIR)
Output Channels	3 (WT, TC, ET)
Initial Filters	32
ROI Size	$160 \times 160 \times 160$ voxels
Dropout Probability	0.1 (regularization)

Table 3: SegResNet configuration for full dataset training.

2.4.5 Training Configuration

Parameter	Value
Total Epochs	35
Batch Size	1 (RTX 5880 Ada, 48 GB VRAM)
Optimizer	AdamW
Learning Rate	5×10^{-5}
Weight Decay	1×10^{-5}
LR Scheduler	Cosine Annealing (cosine decay over epochs)
Loss Function	DiceFocalLoss
- Dice Weight	1.0
- Focal Weight	1.0
- Focal Gamma	2.0 (hard example focus)
Data Augmentation	RandFlip, RandRotate90, RandScaleIntensity, RandShiftIntensity, RandGaussianNoise
Mixed Precision	Enabled (Automatic Mixed Precision)

Table 4: SegResNet training hyperparameters for BraTS 2023 full dataset.

Loss Function Justification:

The DiceFocalLoss combines two complementary objectives:

$$\text{DiceFocalLoss} = \text{Dice Loss} + \text{Focal Loss}$$

- **Dice Loss:** $1 - \frac{2 \times |P \cap G|}{|P| + |G|}$ (primary overlap metric)
- **Focal Loss:** $-\alpha_t (1 - p_t)^\gamma \log(p_t)$ (hard example mining with $\gamma = 2.0$)

The focal component is critical because Enhancing Tumor (ET) comprises only ~5% of voxels, making it easily overlooked by standard loss functions. Focal loss upweights misclassifications on ET, balancing class imbalance.

2.5 Complete Pipeline Workflow

2.5.1 Data Flow Diagram

The complete pipeline processes data through seven sequential stages:

Figure 2: Complete data processing pipeline from raw BraTS data to XAI outputs and VR visualization.

2.5.2 Stage 1: Data Acquisition & Preprocessing

Input: BraTS 2023 raw NIfTI files (4 modalities $\times$ ~1,250 patients)

Processing Steps:

1. **Load MRI Modalities:** Read T1, T1-contrast, T2, T2-FLAIR from NIfTI format
2. **Foreground Cropping:** Remove empty space around brain (reduces volume from ~250 to ~160 voxels per dimension)
3. **Channel-First Formatting:** Reshape to [4, 160, 160, 160] (PyTorch convention)
4. **Per-Channel Z-Score Normalization:**

$$x_{\text{norm}} = \frac{x - \mu_{\text{non-zero}}}{\sigma_{\text{non-zero}}}$$

Computed only on non-zero voxels to preserve background structure

5. **Label Encoding (Training Only):**

- Raw labels: {0=background, 1=necrosis, 2=edema, 3=enhancing}
- Encode into 3 binary channels: WT, TC, ET

Output: Normalized [4, 160, 160, 160] volumes ready for model input

2.5.3 Stage 2: Model Training

Training Loop (35 epochs):

For each epoch:

1. Iterate over training DataLoader (batch size = 1)
2. Forward pass: $y = \text{SegResNet}(x)$
3. Compute DiceFocalLoss
4. Backward pass with mixed precision
5. Update weights via AdamW
6. Log metrics to Weights & Biases

Validation: After each epoch, evaluate on 125 validation patients using Dice, HD95, IoU, Sensitivity, Specificity

Best Model Checkpoint: Save model with highest validation Dice score

2.5.4 Stage 3: Test Set Segmentation

Input: ~125 test patients (never seen during training)

Sliding Window Inference:

- Due to GPU memory constraints, inference uses a sliding window approach
- Window size: 160^3 , stride: $\sim 80^3$ (50% overlap)
- Overlap regions averaged for smooth predictions

Output: Predicted segmentation masks [3, 160, 160, 160] for each test patient

2.5.5 Stage 4: XAI Method Generation (Parallel)

For each test patient and each XAI method:

Grad-CAM:

1. Forward pass through SegResNet
2. Hook into bottleneck layer (256 channels, 20^3)

3. Compute gradients for target class
4. Global average pool $\rightarrow \alpha_k^c$ weights
5. Weighted sum + ReLU + normalize
6. Trilinear upsample $20^3 \rightarrow 160^3$
7. Export as NIfTI

Guided Backpropagation:

1. Register ReLU backward hooks (positive gate only)
2. Backward pass for target class score
3. Extract input-layer gradients
4. Take max absolute gradient across 4 channels
5. Normalize to $[0, 1]$
6. Export as NIfTI

Guided Grad-CAM:

1. Generate Grad-CAM (as above)
2. Generate Guided Backprop (as above)
3. Element-wise multiply + renormalize
4. Export as NIfTI

LRP (Input $\times$ Gradient):

1. Compute gradient of score w.r.t. input: $\partial \text{Score} / \partial x$
2. Multiply element-wise: $x \odot (\partial \text{Score} / \partial x)$
3. Take max across channels
4. Normalize to $[0, 1]$
5. Export as NIfTI

Occlusion Sensitivity:

1. Compute baseline score: $f_c(x)$
2. For each (i,j,k) position (stride=8):
 - a. Mask a 16^3 region centered at (i,j,k) to 0
 - b. Forward pass: $f_c(x_{\text{occluded}})$
 - c. Compute drop: $S_c(i,j,k) = f_c(x) - f_c(x_{\text{occluded}})$
3. Result: 20^3 sensitivity map
4. Trilinear upsample $\rightarrow 160^3$
5. Export as NIfTI

MC Dropout:

1. Set model to eval mode, re-enable Dropout layers only
2. For n in range(20):
 - a. Forward pass with dropout stochasticity
 - b. Apply sigmoid $\rightarrow$ probability $\in [0, 1]$
 - c. Store prediction
3. Mean: $\bar{p} = \text{mean}(\text{all 20 predictions})$
4. Variance: $\sigma^2 = \text{var}(\text{all 20 predictions})$
5. Export both as NIfTI (mean + uncertainty)

Output: 3 NIfTI files per method (WT, TC, ET) per patient

2.5.6 Stage 5: XAI Evaluation Against Ground Truth

For each patient, saliency map, and ground truth tumor mask:

Metric 1: Pointing Game

$$\text{PG} = \mathbb{1}(G[\text{argmax}(S)] = 1)$$

Does the peak saliency voxel fall inside the tumor? Binary outcome.

Metric 2: Saliency Coverage

$$\text{Coverage} = \frac{\sum(S \odot G)}{\sum(S)}$$

What fraction of total saliency mass falls inside the tumor?

Metric 3: Saliency IoU

$$\text{IoU} = \frac{|S_{\text{binary}} \cap G|}{|S_{\text{binary}} \cup G|}$$

Threshold saliency at 0.5, compute overlap with GT mask.

Metric 4: Weighted Dice

$$\text{Weighted Dice} = \frac{2 \times \sum(S \times G)}{\sum(S) + \sum(G)}$$

Soft Dice treating continuous saliency as soft prediction weights.

Output: CSV with 4 metrics $\times$ 3 classes $\times$ 125 patients

2.5.7 Stage 6: Visualization in 3D Slicer

All outputs (predictions, ground truth, saliency maps) are exported as NIfTI files loadable in 3D Slicer:

Directory Structure:

```
slicer_export/  
├── Model_Test_Results/  
│   ├── BraTS-GLI-00001-000/  
│   │   ├── patient_mri_t1.nii.gz  
│   │   ├── patient_mri_t2.nii.gz  
│   │   ├── patient_gt.nii.gz  
│   │   └── patient_pred.nii.gz  
│   └── ...  
├── XAI/  
├── Grad_CAM/  
├── GBP/  
├── Guided_Grad_CAM/  
├── LRP/  
├── Occlusion/  
└── MC_Dropout/
```

2.5.8 Stage 7: VR Integration (In Progress)

The final stage (not yet implemented) integrates explainability maps into an immersive VR environment using SlicerVR. This enables clinicians to:

- Rotate and zoom 3D tumor volumes
 - Toggle XAI overlays on/off
 - Compare multiple explanation methods side-by-side
 - Inspect uncertainty regions interactively
-

2.6 Evaluation Summary: Quantitative Assessment Strategy

2.6.1 Model Evaluation Metrics (Segmentation Performance)

The SegResNet model is evaluated on the test set (~125 patients) using five standard medical imaging metrics:

2.6.1.1 Dice Score (Sørensen–Dice Coefficient)

$$\text{Dice}(P, G) = \frac{2 \times |P \cap G|}{|P| + |G|}$$

Interpretation:

- Range: $[0, 1]$
- 1.0 = Perfect overlap
- 0.5 = 50% overlap
- Used as primary BraTS benchmark metric

Per-Class Interpretation:

- **WT Dice:** Overall tumor detection (includes all three sub-regions)
- **TC Dice:** Necrotic core + enhancing region detection
- **ET Dice:** Specific enhancing tumor boundaries (most clinically challenging)

2.6.1.2 Hausdorff Distance 95th Percentile (HD95)

$$\text{HD95}(P, G) = \max(d_{95}(\partial P, \partial G), d_{95}(\partial G, \partial P))$$

Where d_{95} is the 95th percentile of minimum distances from surface voxels.

Interpretation:

- Range: $[0, \infty)$ mm
- Lower is better
- Measures worst-case surface boundary error (in mm)
- More sensitive to boundary accuracy than Dice

2.6.1.3 Jaccard Index (IoU)

$$\text{IoU}(P, G) = \frac{|P \cap G|}{|P \cup G|}$$

Relationship to Dice: $\text{IoU} = \frac{\text{Dice}}{2 - \text{Dice}}$

Interpretation: Stricter than Dice; penalizes false positives more heavily

2.6.1.4 Sensitivity (Recall / True Positive Rate)

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

Interpretation:

- Range: [0, 1]
- High sensitivity = model detects most tumor voxels
- Important for clinical safety (avoid missing tumors)

2.6.1.5 Specificity (True Negative Rate)

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Interpretation:

- Range: [0, 1]
- High specificity = model avoids false alarms in healthy tissue
- Important for avoiding over-segmentation

Table 2.2: Expected Performance Targets (BraTS Benchmark)

Metric	WT	TC	ET
Dice Score	0.90+	0.85+	0.78+
HD95 (mm)	5.0–	7.0–	10.0–
IoU	0.82+	0.74+	0.64+
Sensitivity	0.90+	0.80+	0.75+
Specificity	0.99+	0.99+	0.99+

Table 5: Target performance metrics for SegResNet on BraTS 2023 test set.

2.6.2 XAI Evaluation Metrics (Explanation Validity)

Unlike model evaluation metrics (which compare predictions to ground truth), XAI evaluation metrics compare saliency maps to ground truth, answering: **"Is the model's reasoning correct?"**

2.6.2.1 Pointing Game (Localization Accuracy)

$$\text{Pointing Game}(S, G) = \mathbb{1}(G[\text{argmax}(S)] = 1)$$

Definition: Does the single voxel with highest saliency fall inside the tumor?

Computation:

1. Find $(d_{\max}, h_{\max}, w_{\max}) = \text{argmax of saliency map } S$
2. Check: $G[d_{\max}, h_{\max}, w_{\max}] \stackrel{?}{=} 1$
3. Return 1 (hit) or 0 (miss)

Aggregation: Accuracy = (Number of hits) / (Total patients) $\times$ 100%

Interpretation:

- 100% = Model always focuses peak attention on tumor
- 50% = Random chance (model confused)
- 0% = Model looks only at non-tumor regions (fundamental failure)

Example:

Patient A: Saliency peak at tumor voxel $\rightarrow$ 1 (hit)

Patient B: Saliency peak at background voxel $\rightarrow$ 0 (miss)

Accuracy = $(1 + 0) / 2 = 50\%$

2.6.2.2 Saliency Coverage

$$\text{Coverage}(S, G) = \frac{\sum(S \odot G)}{\sum(S)}$$

Definition: What fraction of the model's total attention is focused on the tumor?

Computation:

1. Element-wise multiply saliency $\times$ GT mask $\rightarrow$ keeps only saliency inside tumor
2. Sum: $\text{attention_on_tumor} = \sum(S \times G)$
3. Sum all saliency: $\text{total_attention} = \sum(S)$
4. Coverage = $\text{attention_on_tumor} / \text{total_attention}$

Interpretation:

- 1.0 = 100% of attention on tumor (perfect focus)
- 0.75 = 75% on tumor, 25% on background
- 0.5 = Balanced attention (not good)
- 0.0 = No attention on tumor (failure)

Advantage: Independent of saliency shape; only cares about mass distribution

Example:

Saliency: [0.9, 0.8, 0.5, 0.3, 0.1, 0.0] (6 voxels)

GT: [1, 1, 1, 0, 0, 0] (tumor in first 3 voxels)

Sum inside: $0.9 + 0.8 + 0.5 = 2.2$

Sum total: $0.9 + 0.8 + 0.5 + 0.3 + 0.1 = 2.6$

Coverage = $2.2 / 2.6 \approx 0.85$ (85%)

2.6.2.3 Saliency IoU (Intersection over Union)

$$\text{Saliency IoU}(S, G, \tau) = \frac{|S_{\text{binary}} \cap G|}{|S_{\text{binary}} \cup G|}$$

Where $S_{\text{binary}}(i) = \mathbb{1}(S(i) \geq \tau)$ with threshold $\tau = 0.5$.

Definition: How well does the shape of high-saliency regions match the tumor shape?

Computation:

1. Threshold saliency at 0.5 $\rightarrow$ binary "attention mask"
2. Compute intersection: voxels in both S_{binary} and G
3. Compute union: voxels in either S_{binary} or G (or both)
4. IoU = intersection / union

Interpretation:

- 1.0 = Perfect shape overlap
- 0.5 = Moderate overlap
- 0.0 = No overlap

Why Grad-CAM Scores Low:

Grad-CAM produces a diffuse blob due to upsampling from 20^3 . After thresholding, only the blob's core (small region) remains, leading to small intersection relative to union.

Example:

S_binary: [1, 1, 0, 0, 0, 0, 0, 0] (thresholded attention)

GT: [1, 1, 1, 0, 0, 0, 0, 0] (tumor boundary extends further)

Intersection: [1, 1, 0, ...] → 2 voxels

Union: [1, 1, 1, ...] → 3 voxels

IoU = $2 / 3 \approx 0.67$

2.6.2.4 Weighted Dice (Novel Metric)

$$\text{Weighted Dice}(S, G) = \frac{2 \times \sum(S \times G)}{\sum(S) + \sum(G)}$$

Definition: Soft Dice treating continuous saliency as soft prediction weights

Motivation: Standard Dice compares binary masks. For continuous saliency, a soft variant better captures gradual alignment.

Computation:

1. Element-wise multiply: $S \times G$ (only active inside tumor)
2. Sum products: $\sum(S \times G)$ (soft overlap)
3. Sum saliency: $\sum(S)$ (total attention mass)
4. Sum GT: $\sum(G)$ (tumor volume in voxels)
5. Weighted Dice = $2 \times \text{soft_overlap} / (\text{saliency_mass} + \text{tumor_volume})$

Interpretation:

- 1.0 = Saliency perfectly concentrated on tumor boundary

- 0.5 = Moderate concentration
- 0.0 = No saliency on tumor

Advantage over IoU: Captures intensity information; high saliency inside tumor contributes more than low saliency

Example:

Saliency (S): [0.9, 0.7, 0.0, 0.0]

GT (G): [1.0, 1.0, 0.0, 0.0]

Soft overlap: $0.9 \times 1.0 + 0.7 \times 1.0 = 1.6$

Sum S: 1.6

Sum G: 2.0

Weighted Dice = $(2 \times 1.6) / (1.6 + 2.0) = 3.2 / 3.6 \approx 0.89$

2.6.3 MC Dropout Uncertainty Metrics

For MC Dropout, additional metrics assess the quality of uncertainty estimates:

2.6.3.1 Uncertainty Area Ratio (UAR)

$$\text{UAR} = \frac{\sum(\sigma^2 \odot G)}{\sum(\sigma^2)}$$

Definition: What fraction of total uncertainty mass falls inside the tumor?

Interpretation: High UAR = uncertainty concentrated on relevant regions (good)

2.6.3.2 Boundary Uncertainty Ratio

$$\text{Boundary Ratio} = \frac{\sum(\sigma^2 \odot B)}{\sum(\sigma^2)}$$

Where B is a dilated band around the GT mask boundary (± 2 voxels).

Definition: What fraction of uncertainty falls at the tumor boundary (expected behavior)?

Interpretation: High ratio = "boundary glow" effect (uncertainty properly concentrated at edges, indicating model confidence inside tumor)

2.6.3.3 Mean Uncertainty (Inside/Outside)

$$\mu_{\text{inside}} = \frac{\sum(\sigma^2 \odot G)}{|G|} \quad , \quad \mu_{\text{outside}} = \frac{\sum(\sigma^2 \odot (1 - G))}{|(1 - G)|}$$

Interpretation:

- Low μ_{inside} = Confident inside tumor (good)
- Low μ_{outside} = Confident outside tumor (good)
- High both = Generally uncertain (brittle model)

2.6.4 Evaluation Results Organization

All metrics are computed and organized in CSV files:

Evaluation Type	Filename	Scope
Model Performance	model_test_metrics.csv	Dice, HD95, IoU, Sensitivity, Specificity
Per-Patient Model	per_patient_metrics.csv	Model metrics broken down by patient
Grad-CAM XAI	xai_gradcam_metrics.csv	PG, Coverage, IoU, Weighted Dice (Grad-CAM)
GBP XAI	xai_gbp_metrics.csv	Same 4 metrics (GBP)
Guided Grad-CAM XAI	xai_guided_gradcam_metrics.csv	Same 4 metrics (Guided Grad-CAM)
LRP XAI	xai_lrp_metrics.csv	Same 4 metrics (LRP)
Occlusion XAI	xai_occlusion_metrics.csv	PG/MSR, Coverage, IoU, Weighted Dice (Occlusion)
MC Dropout	xai_mc_dropout_metrics.csv	Uncertainty metrics (UAR, boundary ratio, etc.)

Table 6: XAI and model evaluation CSV outputs.

2.6.5 Statistical Aggregation

For each XAI method and metric, summary statistics are computed across all 125 test patients:

Statistic	Computation
Mean	$\mu = \frac{1}{N} \sum_{i=1}^N m_i$
Std Dev	$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (m_i - \mu)^2}$
Min / Max	$\min(m_i), \max(m_i)$
Median	50th percentile
Q1 / Q3	25th and 75th percentiles

Table 7: Statistical aggregation for XAI metrics across test population.

Example Summary (Hypothetical):

Grad-CAM — Pointing Game Accuracy (WT):

Mean: 85.6%

Std Dev: 12.3%

Median: 90.0%

Min: 40.0% (Patient 00291 — exception case)

Max: 100.0%

2.7 Summary of Methodology

This chapter has presented a comprehensive, modular framework for generating and evaluating explainability in 3D brain tumor segmentation models:

Key Contributions:

1. **Six XAI Methods:** Implemented Grad-CAM, Guided Backprop, Guided Grad-CAM, LRP, Occlusion, and MC Dropout adapted for 3D volumetric data
2. **Quantitative Evaluation:** Four novel XAI metrics (Pointing Game, Coverage, IoU, Weighted Dice) provide objective assessment of explanation quality
3. **Uncertainty Quantification:** MC Dropout provides per-voxel confidence estimates enabling identification of brittle decision regions
4. **Complete Evaluation Pipeline:** Integrated model performance (Dice, HD95) with explanation validity (XAI metrics)

5. VR-Ready Outputs: All explanations exported as NIfTI files for visualization in clinical-grade 3D Slicer and immersive VR environments

Alignment with Dissertation Goals:

- ✓ Removes black box problem through six parallel XAI techniques
- ✓ Demonstrates end-to-end transparency from input to decision
- ✓ Provides quantitative metrics validating that explanations match clinical reality
- ✓ Enables VR visualization for immersive clinical evaluation
- ✓ Escapes 2D screening limitations through volumetric 3D analysis

The following chapter (Chapter 3) presents the quantitative results of applying this framework to the BraTS 2023 test set, demonstrating which XAI methods most effectively explain SegResNet decisions and how well they align with ground truth tumors.

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